

# Dr. Jawahar Singh

Plant Molecular Biologist · Legume–Microbe Symbiosis & Nitrogen Fixation  
Postdoctoral Research Associate, Sainsbury Laboratory, University of Cambridge, UK

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## ABOUT / RESEARCH SUMMARY

*Unravelling Symbiotic Interactions in Legumes for Sustainable Crop Productivity.* I decode the molecular and genetic mechanisms governing root nodule symbiosis, nutrient signalling, and stress adaptation in legumes to enhance biological nitrogen fixation (BNF). My long-term goal is to build a research programme that decodes the cell-type-specific nutrient-signalling logic of nodulation and rewires it through genome engineering to generate climate-resilient, low-fertilizer legume varieties in cowpea, chickpea, and soybean - priority crops for food security in South Asia.

## RESEARCH & ACADEMIC INTERESTS

- Root nodule symbiosis and biological nitrogen fixation** - signaling, development, and cell biology of the legume–rhizobium partnership.
- Nutrient-signalling control of symbiosis** - phosphate, nitrate, and iron regulation of nodule development.
- Single-cell & regulatory genomics** - snRNA-seq atlases, gene-regulatory-network inference, and comparative legume genomics.
- Genome engineering & abiotic stress adaptation** - CRISPR-based rewiring and stress-resilient crop improvement.

## RESEARCH APPROACHES

Single-nucleus transcriptomics, gene-regulatory-network inference and comparative/evolutionary genomics across nodulating and non-nodulating legumes to identify the regulatory logic of nodule development; CRISPR-Cas9 and promoter engineering in cowpea and soybean, with hairy-root and stable transformation for functional testing; protein–DNA and protein–protein biochemistry (EMSA, ChIP-seq, Y2H, BiFC) and confocal cell biology to resolve mechanism.

## EDUCATIONAL QUALIFICATIONS

| Year      | Degree        | Discipline / Specialization                | University / Institute                                                |
|-----------|---------------|--------------------------------------------|-----------------------------------------------------------------------|
| 2016–2022 | Ph.D.         | Plant Molecular Biology / Legume Symbiosis | National Institute of Plant Genome Research (NIPGR), New Delhi, India |
| 2012–2014 | M.Sc.         | Biochemistry                               | West Bengal State University, Kolkata, West Bengal, India             |
| 2009-2012 | B.Sc. (Hons.) | Chemistry                                  | West Bengal State University, Kolkata, West Bengal, India             |

## POSITIONS HELD / RESEARCH EXPERIENCE

| Period       | Position                           | Organization                                                                  |
|--------------|------------------------------------|-------------------------------------------------------------------------------|
| 2023–Present | Postdoctoral Research Associate    | Sainsbury Laboratory, University of Cambridge, UK · Prof. Sebastian Schornack |
| 2022–2023    | DGAPA Postdoctoral Fellow          | FES Iztacala, UNAM, Mexico · Prof. Oswaldo Valdés-López                       |
| 2016–2022    | Doctoral Researcher (CSIR-SRF/JRF) | NIPGR, New Delhi, India · Prof. Praveen K. Verma                              |
| 2015–2016    | Research Fellow                    | Jai Narain Vyas University, Jodhpur, India · Prof. Shweta Jha                 |

## KEY RESEARCH CONTRIBUTIONS

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- **University of Cambridge, UK:** Developing multi-omics strategies to identify regulatory hubs controlling nodule development and senescence in cowpea, targeting improved N-fixation efficiency under field conditions.
- **UNAM, Mexico:** First to demonstrate that phosphate signalling regulates the master nodulation gene NIN; identified a PHR-like transcription factor binding the NIN promoter, linking phosphate availability to symbiotic regulation.
- **NIPGR, India:** Uncovered MAP kinase as a key downstream component of Nod-factor perception in chickpea and characterized LysM-RLK receptors in root nodule formation.
- **JNVU, India:** Established proteomic and physiological markers for salinity-stress tolerance in pearl millet, contributing to stress-resilient crop development.

## PUBLICATIONS

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~35 peer-reviewed publications (journal articles, commentaries, and book chapters). \* denotes corresponding / co-corresponding author.

### A. Selected / Key Journal Articles

1. Singh, J., Valdés-López, O., & Schornack, S. (2026). Beyond Nitrogen: Phosphate Controls Root Nodule Symbiosis Commitment. **Trends in Plant Science**.
2. Singh, J.\*, Mendoza-Soto, A. B., Tiwari, M., Acevedo-Sandoval, T. T., Formey, D., Ané, J. M., Isidra-Arellano, M. C., & Valdés-López, O. (2025). Phosphate deficiency reduces nodule formation through a phosphate starvation response-like protein in *Phaseolus vulgaris*. **Plant and Cell Physiology**.
3. Singh, J., Isidra-Arellano, M. C., & Valdés-López, O. (2023). Harnessing the potential of symbiotic associations of plants in nutrient-deficient soil for sustainable agriculture. **Plant and Cell Physiology**.
4. Singh, J.\*, & Verma, P. K. (2023). Role of Nod factor receptors and its allies involved in nitrogen fixation. **Planta**.
5. Singh, J.\*, Varshney, V., & Jha, S. (2023). Genome-wide identification and expression analysis of glycogen synthase kinase encoding genes in foxtail millet (*Setaria italica* L.) under salinity, dehydration, and oxidative stress. **Plant Stress**.
6. Singh, J., & Valdés-López, O. (2022). Discovering the genetic modules controlling root nodule symbiosis under abiotic stresses: salinity as a case study. **New Phytologist**.
7. Singh, J., & Verma, P. K. (2022). Genome-wide identification, expression, and characterization of CaLysM-RLKs in chickpea root nodule symbiosis. **Environmental and Experimental Botany**.
8. Singh, J., & Valdés-López, O. (2022). A nodule peptide confiscates haem to promote iron uptake in rhizobia. **Trends in Plant Science**.
9. Singh, J., & Verma, P. K. (2021). NSP1 allies with GSK3 to inhibit nodule symbiosis. **Trends in Plant Science**.

### B. Other Journal Articles

1. Jha, S.\*, Tyagi, S., Soni, S., Choudhary, D., & Singh, J.\* (2025). The role of PIN-FORMED (PIN) auxin efflux transporters in plant abiotic stress responses. **Physiologia Plantarum**.
2. Li, F., Singh, J.\*, & Kumar, A.\* (2025). Symbiotic Advantage Beyond Root Architecture: How AMF Levels the Playing Field for Phosphorus Uptake in Sorghum. **Molecular Plant-Microbe Interactions**.
3. Singh, J., Mishra, V., & Varshney, V. (2024). Arsenic tolerance unveiled in Arabidopsis: CPK23 and PHT1;1 alliance. **Journal of Plant Biochemistry and Biotechnology**.
4. Shukla, K., Mishra, V., Singh, J., Varshney, V., Verma, R., & Srivastava, S. (2024). Nanotechnology in sustainable agriculture: A double-edged sword. **Journal of the Science of Food and Agriculture**.
5. Singh, J.\*, Mishra, V., Varshney, V., & Jha, S. (2023). Interplay of calcium signaling and ERF-VII stability in plant hypoxia tolerance. **Functional & Integrative Genomics**.
6. Singh, J.\*, Varshney, V., & Mishra, V. (2023). AUR1 and its pals: orchestration of intracellular rhizobia infection in legume for nitrogen fixation. **Plant Cell Reports**.

7. Pasari, N., Gazara, R. K., Singh, J., & Verma, P. K. (2023). Genome-Wide Identification of *SWEET* Genes in *Cicer arietinum* and Modulation of Its Expression in Endophytic Interactions with *Serendipita indica*. **Journal of Plant Growth Regulation**.
8. Varshney, V., Singh, J., & Mishra, V. (2023). Unlocking the plant ER stress code: IRE1-proteasome signaling cohort takes the lead. **Trends in Plant Science**.
9. Isidra-Arellano, M. C., Singh, J., & Valdés-López, O. (2023). Unraveling the potential of the strigolactones-NSP1/NSP2 friendship in crop improvement. **Trends in Plant Science**.
10. Jha, S., Singh, J., Chouhan, C., Singh, O., & Srivastava, R. K. (2022). Evaluation of multiple salinity tolerance indices for screening and comparative biochemical and molecular analysis of pearl millet genotypes. **Journal of Plant Growth Regulation**.
11. Jha, S., Maity, S., Singh, J., Chouhan, C., Tak, N., & Ambatipudi, K. (2022). Integrated physiological and comparative proteomics analysis of contrasting genotypes of pearl millet reveals underlying salt-responsive mechanisms. **Physiologia Plantarum**.

### C. Commentaries / Editorials

1. Singh, J.\* (2025). Metabolic detour, symbiotic delay: Insights from *S. meliloti* suppressor mutants. **Molecular Plant-Microbe Interactions**.
2. Prakash, V., & Singh, J. (2025). A viral brake on bloom: BYDV-GAV delays flowering via VOZ degradation. **Molecular Plant-Microbe Interactions**.
3. Tiwari, R.\*, & Singh, J. (2025). Tracking rust effector protein delivery into plant cells. **Molecular Plant-Microbe Interactions**.
4. Gomez-Gutierrez, S. V., & Singh, J. (2025). When Mutualism Turns Parasitic: How Alfalfa Balances Cooperating and Cheating Rhizobia. **Molecular Plant-Microbe Interactions**.
5. Singh, J.\* (2024). Fortifying Plant Armor: CESA3 Enhances *Arabidopsis thaliana*'s Defense Against Bacterial Wilt Under Heat Stress. **Molecular Plant-Microbe Interactions**.
6. Singh, J.\*, & V Prakash, V. (2024). Rethinking Growth and Defense: miR472 as a Sensor and Regulator of Plant Immunity. **Molecular Plant-Microbe Interactions**.
7. Tiwari, R., & Singh, J. (2024). Decoding the Fancy Coat Worn by Rhizobia in Symbiosis. **Molecular Plant-Microbe Interactions**.

### D. Book Chapters

1. Jha, S., Singh, J., Dutta, S., Varshney, V., Mishra, V., Choudhary, D., & Tak, N. (2025). Regulatory genes for the improvement of salt and drought tolerance. In *Exogenous Priming and Engineering of Metabolic and Regulatory Genes*. Elsevier.
2. Mishra, V., Shukla, P., Shukla, K., Varshney, V., Singh, J., & Srivastava, S. (2025). Strategies for Increasing Zinc Uptake Efficiency in Plants. In *Zinc in Soil-Plant Continuum* (pp. 139–161). Springer.
3. Singh, J., Varshney, V., Jha, S., Saadhe, A., Manuka, R., & Penna, S. (2024). Molecular techniques for improvement of abiotic stress tolerance in plants. In *Innovative Methods in Horticultural Crop Improvement*. Springer.
4. Mishra, V., Singh, J., & Varshney, V. (2024). Biotechnological insights into cold-stressed adapted microorganisms for plant health and soil improvement. In *Extremophiles for Sustainable Agriculture and Soil Health Improvement*. Springer.
5. Varshney, V., Singh, J., Dutta, S., Mishra, V., & Jha, S. (2024). Understanding the problem at hand: nanotechnological implications that cannot be foreseen. In *Molecular Impacts of Nanoparticles on Plants and Algae* (pp. 397–424). Academic Press.
6. Singh, J., & Verma, P. K. (2023). Plant transcription factors and nodule development. In *Plant Transcription Factors* (pp. 175–196). Academic Press.
7. Varshney, V., Singh, J., & Salvi, P. (2023). Sugar Signaling and Their Interplay in Mitigating Abiotic Stresses in Plant. In *Smart Plant Breeding for Field Crops in Post-genomics Era*. Springer.
8. Singh, J., Kumar, K., & Verma, P. K. (2020). Functional characterization of genes involved in legume nodulation using hairy root cultures. In *Hairy Root Cultures Based Applications* (pp. 217–228). Springer.

## AWARDS, HONOURS & FELLOWSHIPS

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- **DGAPA Postdoctoral Fellowship (2022–2023)** - DGAPA, Universidad Nacional Autónoma de México (UNAM). Prestigious international postdoctoral fellowship to investigate phosphate regulation of nodulation in legumes.
- **CSIR Senior Research Fellowship (SRF, 2018–2021) & UGC Junior Research Fellowship (JRF, 2016–2018)** - Council of Scientific and Industrial Research / University Grants Commission, Government of India. Competitively awarded to support doctoral research.
- **Swachhta Saarthi Fellowship (2021)** - Waste to Wealth Mission, Office of the Principal Scientific Adviser to the Government of India. National-level young-researcher award for sustainability and science outreach.
- **Graduate Aptitude Test in Engineering (GATE), 2016** - MHRD, Government of India. Qualified with national ranking for Ph.D. admissions in life sciences and biotechnology.

## RESEARCH GUIDANCE & MENTORING

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- **Master's Research Students - University of Cambridge (2025–Present):** Mentored master's research projects on root nodule development and molecular regulation of symbiosis; hands-on training in molecular biology, confocal imaging, and data analysis.
- **Summer Research Interns - University of Cambridge (2023–Present):** Mentored several summer students over 10-week projects on nodule development and symbiosis regulation.
- **Undergraduate Supervision - University of Cambridge (2024–Present):** Supervisor for the Plant and Microbial Sciences (IBPMS) course; revision sessions, essay guidance, and research-data interpretation.

## ACADEMIC SERVICE, EDITORIAL & PEER REVIEW

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- **Assistant Feature Editor** - Molecular Plant-Microbe Interactions (2024–Present). Accessible research summaries, author interviews, and commentaries promoting early-career researcher visibility.
- **Peer Reviewer** - The Plant Cell, New Phytologist, The Plant Journal, Journal of Experimental Botany, Frontiers in Plant Science, BMC Plant Biology, Journal of Genetics, South African Journal of Botany, Scientific Reports.

## SCIENTIFIC EVENT ORGANIZATION & LEADERSHIP

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- Satellite Meeting Co-organizer & Chair - "Building Careers in MPMI through Effective Mentoring", IS-MPMI Congress 2025, Cologne, Germany.
- Poster Judge, Best Poster Award - IS-MPMI Congress 2025, Cologne, Germany.
- Mini-Symposium Co-organizer - "Molecular Mechanisms of Plant Symbiosis and Pathogen Interaction" (21 July 2025), Sainsbury Laboratory, University of Cambridge.
- Scientific Committee Member & Abstract Review Panel - Sainsbury Laboratory Symposium (17–19 September 2024), University of Cambridge.
- Co-convenor - SciEfflux-2018, 10th Annual Student Symposium, NIPGR, India.

## REFEREES

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Prof. Sebastian Schornack (Sainsbury Laboratory, University of Cambridge, UK) · Prof. Oswaldo Valdés-López (FES Iztacala, UNAM, Mexico) · Prof. Praveen K. Verma (School of Life Sciences, JNU, New Delhi, India). Contact details available on request.